

Problem Set 6

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Setup

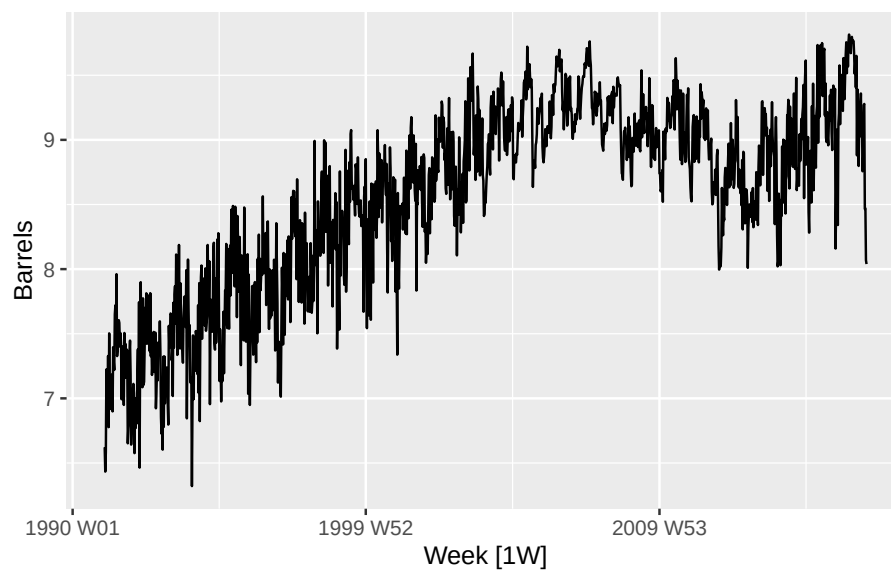
Packages

```
library(fpp3)
library(forecast)
set.seed(280) # Reproducibility
```

View Barrels of US Gasoline

```
df <- us_gasoline

df %>%
  autoplot(Barrels)
```



Create Dynamic Harmonic Regression Model

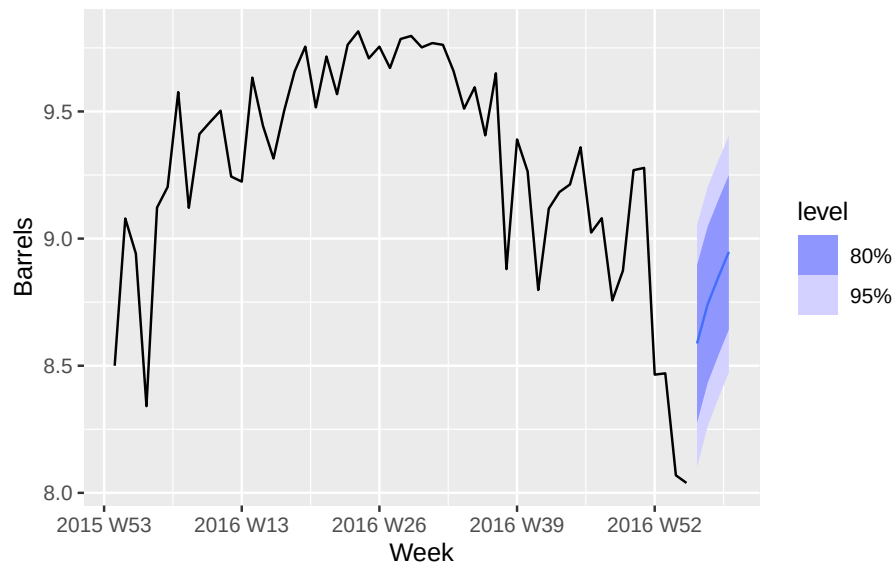
```
lambda <- BoxCox.lambda(df$Barrels)

fit_DHR <- df %>%
  model(ARIMA(box_cox(Barrels, lambda) ~ fourier(K = 5)))
```

Using a BoxCox transformation to get a more stable variance since $\log()$ was insufficient.

```
fc <- forecast(fit_DHR, h = 4)

fc %>%
  autoplot(df %>% filter(as.Date(Week) > as.Date("2016-01-01")))
```

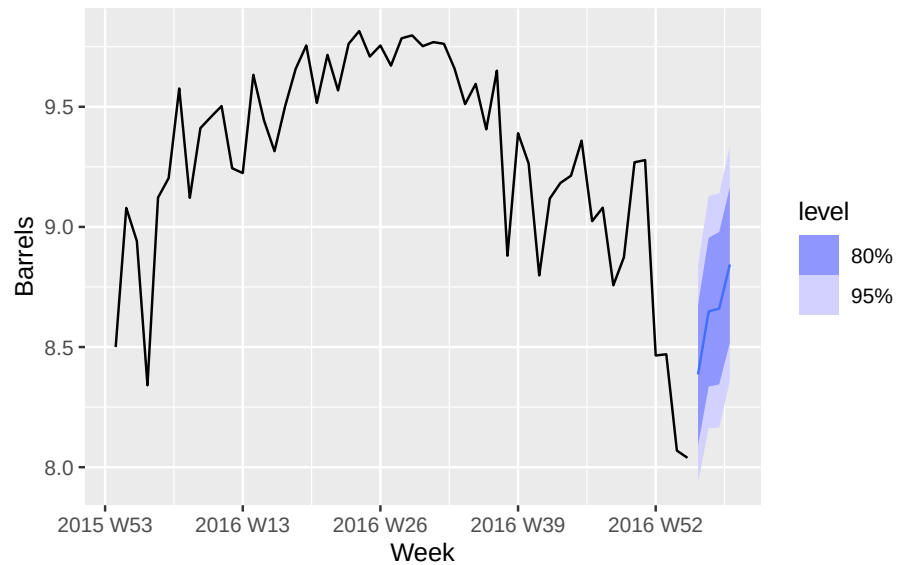


Create Neural Network Model

```
fit_NNETAR <- df %>%
  model(NNETAR(Barrels))

fc <- forecast(fit_NNETAR, h = 4)

fc %>%
  autoplot(df %>% filter(as.Date(Week) > as.Date("2016-01-01")))
```



Neural Network Model:

- NNAR(11,1,6)[52]

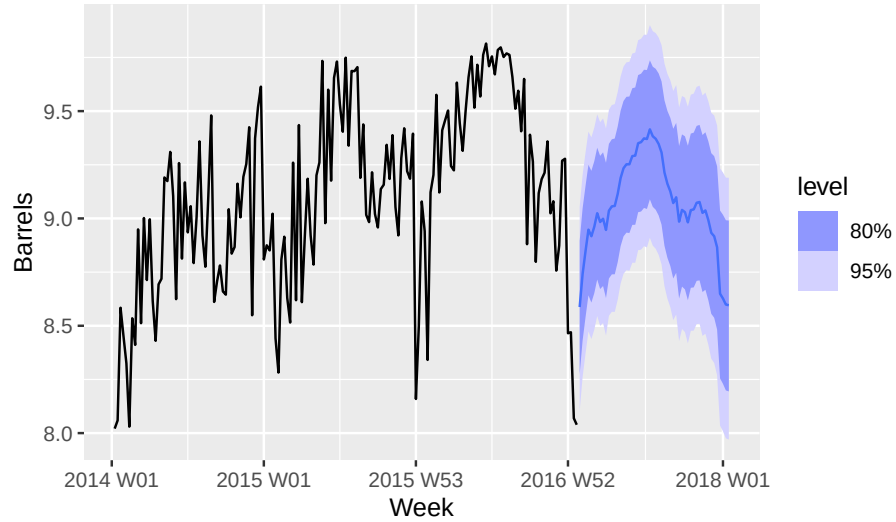
Question 1

Part A

```
fc_DHR <- forecast(fit_DHR, h = 52)

fc_DHR %>%
  autoplot(df %>% filter(as.Date(Week) > as.Date("2014-01-01"))) +
  labs(title = "Dynamic Harmonic Regression Model Forecast")
```

Dynamic Harmonic Regression Model Forecast

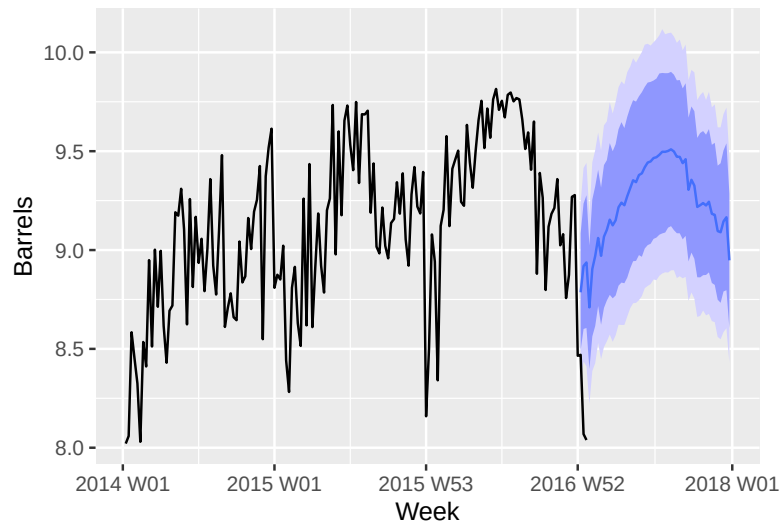


Part B

```
fc_NNETAR <- forecast(fit_NNETAR, h = 52)
```

```
fc_NNETAR %>%  
  autoplot(df %>% filter(as.Date(Week) > as.Date("2014-01-01"))) +  
  labs(title = "Auto Regressive Neural Network Model Forecast")
```

Auto Regressive Neural Network Model Forecast



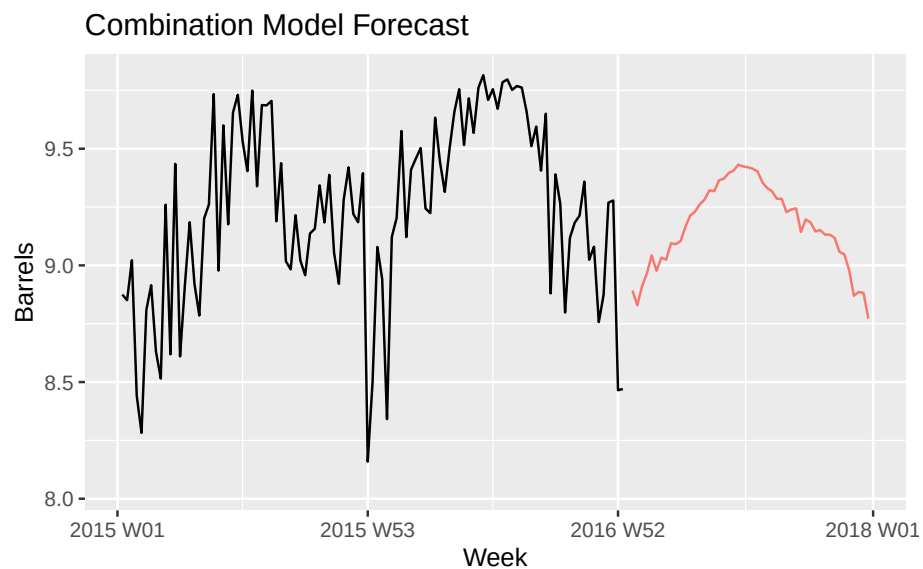
Part C

```
observed <- df %>%
  filter(as.Date(Week) > as.Date("2015-01-01")) %>%
  select(Week, Barrels)

forecasts <- tsibble(
  Week = fc_NNETAR$Week,
  NNETAR = fc_NNETAR$.mean,
  DHR = fc_DHR$.mean,
) %>%
mutate(
  Combination = (NNETAR + DHR) / 2
)

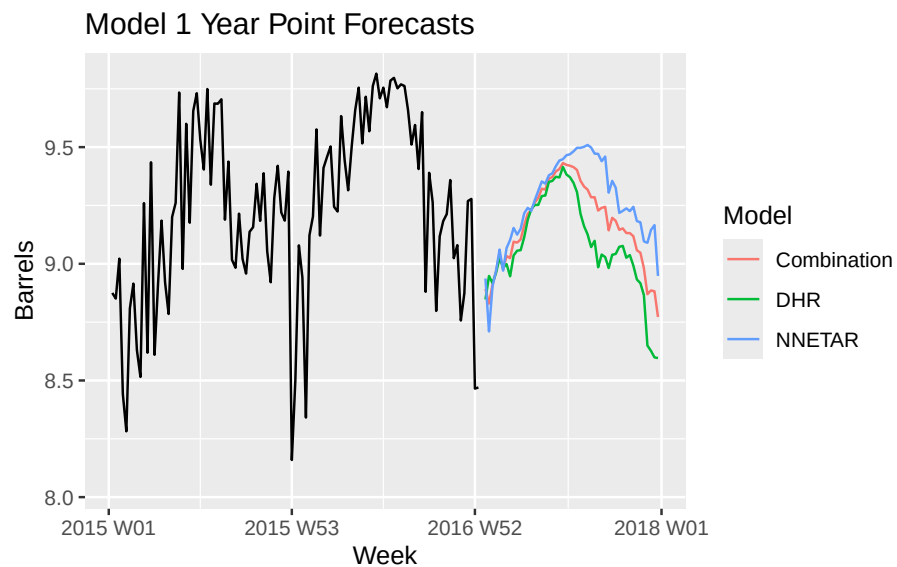
combined <- bind_rows(as_tibble(observed), forecasts)

combined %>%
  ggplot(aes(x = Week)) +
  geom_line(aes(y = Barrels), color = "black") +
  geom_line(aes(y = Combination, color = "red")) +
  labs(
    title = "Combination Model Forecast"
  ) +
  theme(legend.position = "none")
```



Part D

```
combined %>%
  pivot_longer(
    cols = c(NNETAR, DHR, Combination),
    names_to = "Model",
    values_to = "Forecast"
  ) %>%
  ggplot(aes(x = Week)) +
  geom_line(aes(y = Barrels), color = "black") +
  geom_line(aes(y = Forecast, color = Model)) +
  labs(
    title = "Model 1 Year Point Forecasts"
  )
)
```



Part E

```
sim_NNETAR <- fit_NNETAR %>%
  generate(times = 1000, h = 52)

sim_DHR <- fit_DHR %>%
  generate(times = 1000, h = 52)

sim_combined <- sim_NNETAR %>%
  select(Week, .rep, .sim) %>%
  rename(NNETAR = .sim) %>%
```

```

left_join(
  sim_DHR %>% select(Week, .rep, .sim) %>% rename(DHR = .sim),
  by = c("Week", ".rep")
) %>%
mutate(Combination = (NNETAR + DHR)/2)

```

Part F

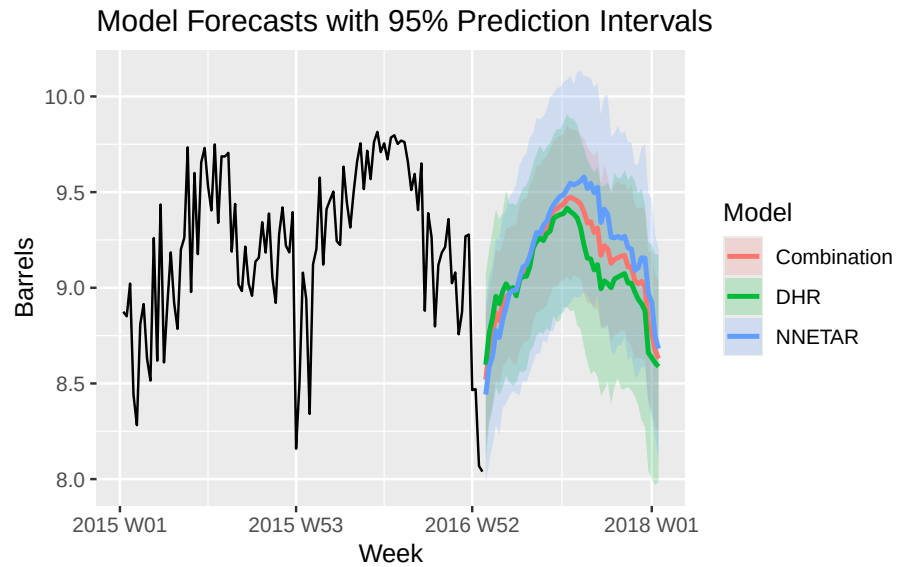
```

# Compute 95% prediction intervals from simulations
pi_df <- sim_combined %>%
  pivot_longer(
    cols = c(NNETAR, DHR, Combination),
    names_to = "Model",
    values_to = "Sim"
  ) %>%
  group_by(Model) %>%
  summarise(
    Lower = quantile(Sim, 0.025),
    Median = quantile(Sim, 0.5),
    Upper = quantile(Sim, 0.975),
    .groups = "drop"
  )

# Plot actual data and prediction intervals
ggplot() +
  # Actuals
  geom_line(
    data = filter(df, as.Date(Week) > as.Date("2015-01-01")),
    aes(x = Week, y = Barrels), color = "black"
  ) +
  # Prediction intervals as ribbons
  geom_ribbon(
    data = pi_df,
    aes(x = Week, ymin = Lower, ymax = Upper, fill = Model),
    alpha = 0.2
  ) +
  # Median forecasts as lines
  geom_line(
    data = pi_df,
    aes(x = Week, y = Median, color = Model),
    size = 1
  ) +
  labs(
    title = "Model Forecasts with 95% Prediction Intervals",
    y = "Barrels",

```

```
x = "Week"
)
```



Assistance utilized:

[<https://chatgpt.com/share/69330866-42a4-8010-bc1c-9e98d330e807>]

Part G

A test set is required to compute the Winkler scores.

The following steps are taken below:

- Limit Training Set as 1991 to 2015 (2016 saved for testing)
- Fit the DHR and NNETAR models on the training data
- Mutate to add Combination
- Create Forecasts
- Generate Sample Paths
- Summarise paths using `dist_sample()` to create prediction intervals
- Calculate Winkler Scores using testing data

```
# Drop 3 Weeks in 2017
df <- us_gasoline %>%
```



```

  filter(year(Week) < 2017)

# Keep the last year as a test set
train <- df %>%
  filter(year(Week) < 2016)

lambda_train <- BoxCox.lambda(train$Barrels)

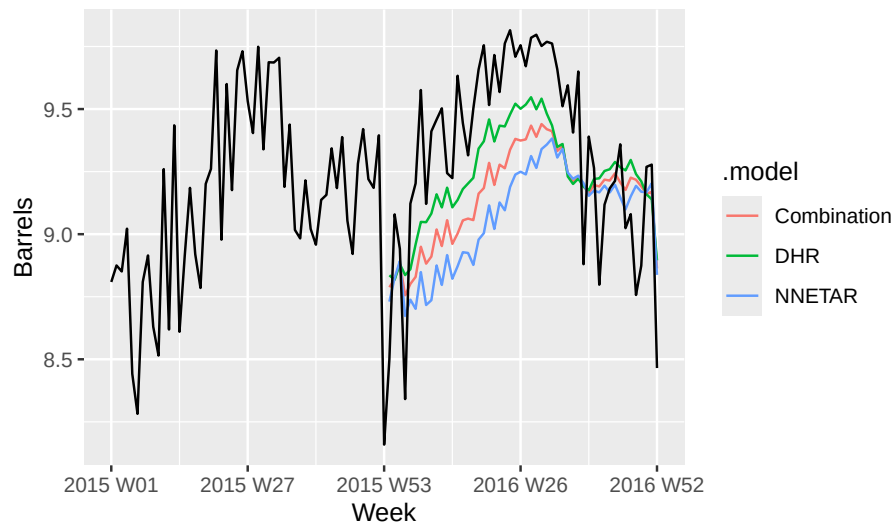
models <- train %>%
  model(
    DHR = ARIMA(box_cox(Barrels, lambda_train) ~ fourier(K = 5)),
    NNETAR = NNETAR(Barrels)
  ) %>%
  mutate(Combination = (DHR + NNETAR) / 2)

fc <- models %>%
  forecast(h = 52)

fc %>%
  autoplot(df %>% filter(year(Week) > 2014), level = NULL) +
  labs(title = "Model Point Forecasts")

```

Model Point Forecasts



```

fc %>%
  accuracy(df) %>%
  arrange(RMSE)

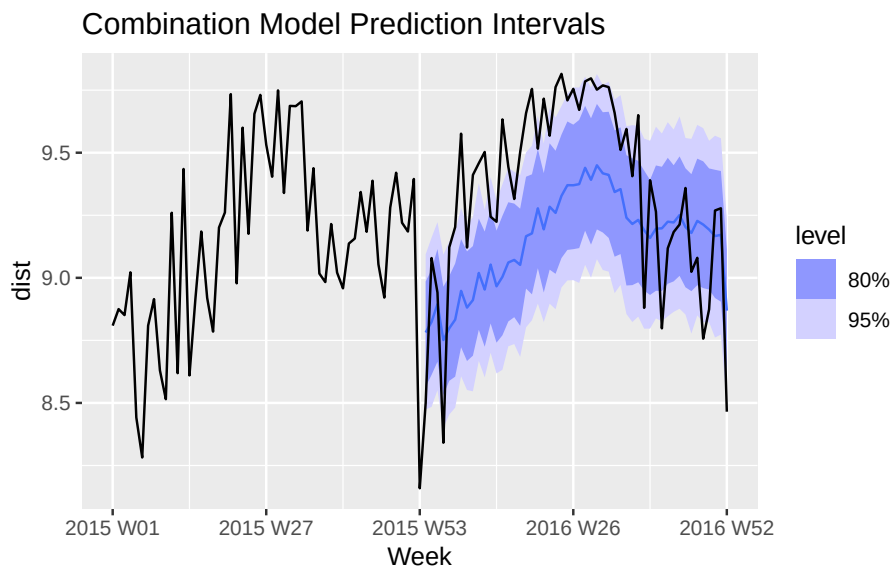
```

A tibble: 3 x 10

	.model	.type	ME	RMSE	MAE	MPE	MAPE	MASE	RMSSE	ACF1
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	DHR	Test	0.118	0.285	0.255	1.17	2.73	0.893	0.792	0.301
2	Combination	Test	0.201	0.352	0.318	2.05	3.39	1.12	0.979	0.457
3	NNETAR	Test	0.285	0.437	0.384	2.93	4.07	1.35	1.21	0.590

```
futures <- models %>%
  generate(h = 52, times = 1000) %>%
  as_tibble() %>%
  group_by(Week, .model) %>%
  summarise(
    dist = distributional::dist_sample(list(.sim))
  ) %>%
  ungroup() %>%
  as_fable(
    index = Week,
    key = .model,
    distribution = dist,
    response = "Barrels"
  )
```

```
futures %>%
  filter(.model == "Combination") %>%
  autoplot(filter(df, year(Week) > 2014)) +
  labs(title = "Combination Model Prediction Intervals")
```



```
futures %>%
  accuracy(df, measures = interval_accuracy_measures, level = 95) %>%
```

```
arrange(winkler)
```

```
# A tibble: 3 x 5  
  .model      .type winkler pinball scaled_pinball  
  <chr>      <chr>   <dbl>   <dbl>         <dbl>  
1 DHR        Test     1.08  0.0391         0.0686  
2 NNETAR     Test     2.30  0.120          0.211  
3 Combination Test     2.45  0.132          0.231
```

All models underpredict initially. The Dynamic Harmonic Regression Model underpredicts the least which gives it the most accurate prediction intervals and the lowest Winkler score.